

FIITJEE Vadodara CENTRE

PINNACLE CLASS-XII(ADVANCED) Paper 1

PHYSICS , MATHEMATICS PHASE – 5 DATE : 30/07/2024

Time Allotted: 3 Hours

Maximum Marks: 182

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section 1** is Physics, **Section 2** is Chemistry and **Section 3** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & C** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Marking Scheme for All Two Parts.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –1 In all other cases.
 For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–1 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-13)** – Contains six (06) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (ii) **Part-C (01-05)** contains five (05) Numerical based questions, the answer of which is an Integer ranging from 0-9 (both inclusive) and each question carries **+3 marks** for correct answer and **there will be no negative marking**.

SECTION - 1:PHYSICS

PART A : (One or More than One Option Correct Type)

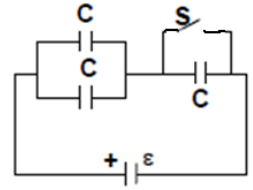
This section contains **7 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONE or More than One option is correct**.

- In the circuit shown, each capacitor has a capacitance C . The emf of the cell is ε . If the switch S is closed,

(A) Some positive charge will flow out of the positive terminal of the cell.
 (B) Some positive charge will enter the positive terminal of the cell.
 (C) the amount of charge flowing through the cell will be $C\varepsilon$.
 (D) the amount of charge flowing through the cell will be $\frac{4}{3}C\varepsilon$.
- Two bulbs consume same power when operated at 200V and 300V respectively. When these bulbs are connected in series across a D.C. source of 500 V, then

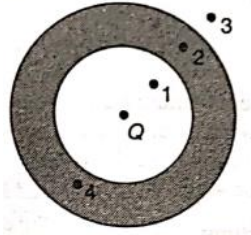
(A) ratio of potential difference across them is $3/2$
 (B) ratio of potential difference across them is $4/9$
 (C) ratio of power consumed across them is $4/9$
 (D) ratio of power consumed across them is $2/3$
- Two equal negative charges $-q$ each are fixed at the points $(0, a)$ and $(0, -a)$ on the y-axis. A positive charge Q is released from rest at the point $(2a, 0)$ on the x-axis. The charge Q will :

(A) Execute simple harmonic motion about the origin
 (B) At origin velocity of particle is maximum.
 (C) Move to infinity
 (D) Execute oscillatory but not simple harmonic motion.

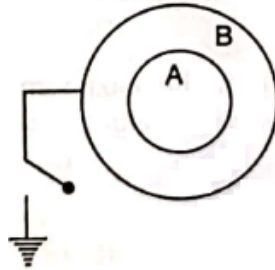


space for rough work

4. At the centre of a conducting thick hollow sphere, as shown in figure, a positive charge Q is placed. Electric field intensity $\vec{E}$ is not be zero at point



- (A) 1 (B) 2 (C) 3 (D) 4
5. If the flux of the electric field through a closed surface is zero, then
 (A) the electric field must be zero everywhere on the surface
 (B) the electric field may be zero everywhere on the surface
 (C) the net charge inside the surface must be zero
 (D) the charge in the vicinity of the surface must be zero
6. Two concentric shells have radii R and $2R$, charges q_A and q_B and potentials $2V$ and $\left(\frac{3}{2}\right)V$, respectively. Now, shell B is earthed and let charges on them become q'_A and q'_B . Then



- (A) $\frac{q_A}{q_B} = \frac{1}{2}$
 (B) $\frac{|q'_A|}{|q'_B|} = 1$
 (C) potential of A after earthing beomes $\left(\frac{3}{2}\right)V$
 (D) potential difference between A and B after earthing becomes $V/2$
7. A particle of mass 2 kg and charge 1 mC is projected vertically with a velocity 10 m/s . There is a uniform horizontal electric field of 10^4 N/C .
 (A) The horizontal range of the particle is 10 m
 (B) The time of flight of the particle is 2 s
 (C) The maximum height reached is 5 m
 (D) The horizontal range of the particle is 0 m

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PART A: (Only One Option Correct Type)

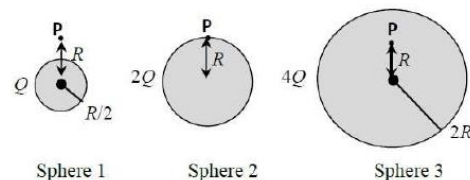
This section contains **6 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE option is correct**.

8. A and B are two points on a uniform ring of resistance R . The $\angle ACB = \theta$, where C is the centre of the ring. The equivalent resistance between A and B is

(A) $\frac{R}{4\pi^2}(2\pi - \theta)\theta$ (B) $R\left(1 - \frac{\theta}{2\pi}\right)$ (C) $R\frac{\theta}{2\pi}$ (D) $R\frac{2\pi - \theta}{4\pi}$

9. Charges Q , $2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$, respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the center of spheres 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then

(A) $E_1 > E_2 > E_3$ (B) $E_3 > E_1 > E_2$
(C) $E_2 > E_1 > E_3$ (D) $E_3 > E_2 > E_1$



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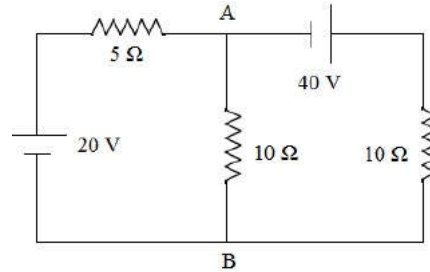
10. At distance of 5 cm and 10 cm outwards from the surface of a uniformly charged solid sphere, the potentials are 100 V and 75 V respectively. Then choose the **incorrect** statement
- (A) Potential at its surface is 150V.
 - (B) The charge on the sphere is $(5/3) \times 10^{-10} \text{ C}$
 - (C) The electric field on the surface is 1500 V/m.
 - (D) The electric field at its center is zero
11. Total charge – Q is uniformly spread along circumference of a fixed ring of radius R. A small test charge +q of mass m is kept at the centre of the ring and is given a gentle push (small displacement) along the axis of the ring. The motion of the charge + q is
- (A) The motion of charged particle is periodic but not SHM
 - (B) The motion of charged particle is SHM with a time period $4\sqrt{\frac{\pi^3 \epsilon_0 m R^3}{Qq}}$
 - (C) The motion of charged particle is SHM with a time period $2\sqrt{\frac{\pi^3 \epsilon_0 m R^3}{Qq}}$
 - (D) The motion of charged particle is SHM with a time period $\sqrt{\frac{\pi^3 \epsilon_0 m R^3}{Qq}}$

space for rough work

12. A solid conducting sphere of radius 10 cm is enclosed by a thin metallic shell of radius 20 cm. A charge $q = 20 \mu\text{C}$ is given to the inner sphere. Find the heat generated in the process, the inner sphere is connected to the shell by a conducting wire (in J).

(A) 5 (B) 6 (C) 9 (D) 8

13. Consider the circuit shown in the figure. The current flowing in the branch AB of the circuit is



(A) 1 A, from A to B (B) 1 A, from B to A
(C) 2 A, from A to B (D) Zero

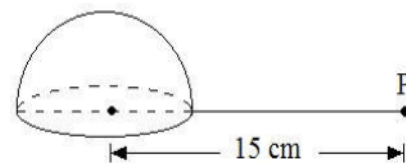
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PART – C: (Integer Value Type)

This section contains 5 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

1. A wire of length L and 3 identical cells of negligible internal resistance are connected in series. Due to the current, temperature of wire rises ΔT in time t . A number of N similar cells is now connected in series with a wire of same material and cross-section but length $2L$. The temperature of wire is raised by same amount ΔT in same time t . Find the value of N

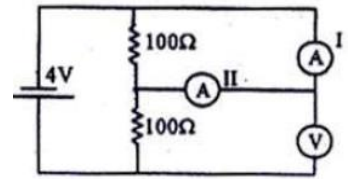
2. Figure shows a solid hemisphere of radius 5 cm, with a charge of 5 nC distributed uniformly through its volume. The hemisphere lies on a plane and a point P is located on the plane, along a radial line from the centre of curvature at a distance 15 cm. The electric potential at P due to hemisphere is $150n$. What is the value of n .



3. The region between two concentric spherical shells of radii R_1 and R_2 includes a volume charge density given by $\rho = \frac{a}{r}$ where a , is a constant and r is the distance from the centre. There is a charge Q located at its centre. The value of ' a ' for which the region $R_1 \leq r \leq R_2$ has a constant electric field intensity is $\frac{Q}{\sqrt{z}\pi R_1^2}$. Find z

space for rough work

4. A 100-W bulb and a 25-W bulb are designed for the same voltage. They have filaments of the same length and material. The ratio of the diameter of the 100-W bulb to that of the 25-W bulb is $x:1$, find the value of x .
5. In the figure, ammeter (I) reads a current of 10 mA, while the volt meter reads a potential difference of 3V. What does ammeter (II) (in mA) read? The ammeters are identical, the internal resistance of the battery is negligible. (consider all ammeters and voltmeters as non ideal) (round off the answer to the nearest integer)



space for rough work

SECTION 2: MATHEMATICS

PART-A: (One or More than One Option Correct Type)

This section contains **7 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **One or More than One option is correct**.

1. If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct options is/are
 (A) $\cos\beta > 0$ (B) $\sin\beta < 0$
 (C) $\cos(\alpha + \beta) > 0$ (D) $\cos\alpha < 0$

 2. Let $f(x) = \max(|x|, \{x\}, \cos x)$, $-\frac{\pi}{2} < x \leq \frac{\pi}{2}$ then :
 (A) $f(0) = 1$ (B) $f'(0) = 0$
 (C) $f(x) = 1$ has 3 solutions (D) $f\left(\frac{\pi}{2}\right) = 0$
-

space for rough work

3. A continuous function $f(x)$ on $R \rightarrow R$ satisfies the relation $f(x) + f(2x + y) + 5xy = f(3x - y) + 2x^2 + 1$ for $\forall x, y \in R$, then which of the following hold(s) good?
- (A) f is many one (B) f has no minima
(C) f is neither odd nor even (D) f is bounded

4. If $f(x) = \frac{\sqrt{x^2 + 2}}{3x - 6}$ then
- (A) $\lim_{x \rightarrow -\infty} f(x) = \frac{-1}{3}$ (B) $\lim_{x \rightarrow \infty} f(x) = \frac{1}{3}$ (C) $\lim_{x \rightarrow -\infty} f(x) = \frac{1}{3}$ (D) $\lim_{x \rightarrow \infty} f(x) = \frac{-1}{3}$

space for rough work

5. If $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}} = e^2$, then the possible values of 'a' & 'b' are
(A) $a = 1, b = 2$ (B) $a = 2, b = 1$ (C) $a = 3, b = 2/3$ (D) $a = 2/3, b = 3$
6. The point at which the function $f(x) = |x - 0.5| + |x - 1| + \tan x$ does not have a derivative in the interval $(0, 2)$ are
(A) 1 (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{1}{2}$
7. Fundamental period of $f(x) = \sec(\sin x)$ is
(A) $\frac{\pi}{2}$ (B) 2π (C) π (D) $\frac{3\pi}{2}$
-

space for rough work

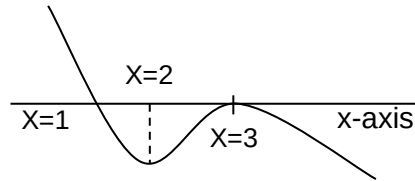
PART - A : (Only One Option Correct Type)

This section contains **6 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE option is correct**.

8. The function $f(x) = a(x^2 - 1)(ax + b)$, ($a \neq 0$) has
 (A) a point of local minima at certain $x \in R^+$
 (B) a point of local minima at certain $x \in R^-$
 (C) a point of local maxima at certain $x \in R^-$
 (D) no point of local maxima / minima
9. The value of $\sum_{r=1}^{10} \sum_{s=1}^{10} \tan^{-1} \frac{r}{s}$ is
 (A) 20π (B) 25π (C) 15π (D) None of these
10. The greatest value of $(3x + 2y)$ subjected to the condition that $x > 0, y > 0$ and $x^2 + xy + y^2 = \frac{27}{7}$, is
 (A) 3 (B) 6 (C) 7 (D) 9
11. $\lim_{x \rightarrow \infty} \left(x - x^2 \ln \left(1 + \frac{1}{x} \right) \right)$ is equal to
 (A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{1}{3}$ (D) 1

space for rough work

12. Range of the function $f(x) = \frac{(x-2)^2}{(x-1)(x-3)}$ is
 (A) $(1, \infty)$ (B) $(-\infty, 1)$ (C) $\mathbb{R} - (0,1]$ (D) $(0,1]$
13. If graph of $y = f'(x)$ is



Then which of following is true for $y = f(x)$

- (A) Point of inflection at $x = 2, x = 3$ (B) Concave down in $(2, 3)$
 (C) Point of local minima at $x = 2$ (D) Point of local maxima at $x = 1$

space for rough work

PART – C: (Integer Value Type)

This section contains 5 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

- Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfy the equation $f(x) = \frac{10-x}{10}$ is _____
- If $\lim_{x \rightarrow 0} (x^{-3} \sin 3x + ax^{-2} + b)$ exists and is equal to zero, then the value of $a + 2b$ is _____
- The domain of the function $y = \sqrt{\sin x + \cos x} + \sqrt{7x - x^2 - 6}$ is $\left[p, \frac{q\pi}{4} \right] \cup \left[\frac{r\pi}{4}, s \right]$ then the value of $(p + q + r + s)$ is $10 +$ _____
- The number of points in $(-\infty, \infty)$, for which $x^4 - x \sin \pi x - \cos^3 \pi x = 0$ is (are) _____
- Number of value(s) of $x \in [-2, 3]$ where $f(x) = |x^2 - x + 2|^3 + \cos|x| + |\cos x|^{5/3} + \{\sin \pi x\} \cdot \sin x \cdot \sin 2x$ is not differentiable is(are) _____ (note : $\{k\}$ denotes the fractional part functions of k)

space for rough work

SECTION – III CHEMISTRY

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A & C** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

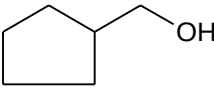
1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

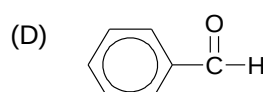
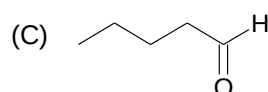
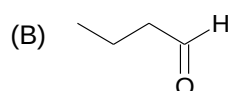
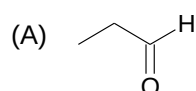
- (i) **PART–A (01–06)** contains six (6) Multiple Choice Questions which have **One or More Than One Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –1 In all other cases.
 For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–1 marks**, as a wrong option is also darkened.
- (ii) **Part-A (07-10)** – This section contains Two (02) List-Match Sets, each List-Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. FOUR options are given in each Multiple Choice Question based On List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question. Each question carries **+3 Marks** for correct combination chosen and **–1 mark** for wrong options chosen.
- (iii) **Part-C (01-08)** contains eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

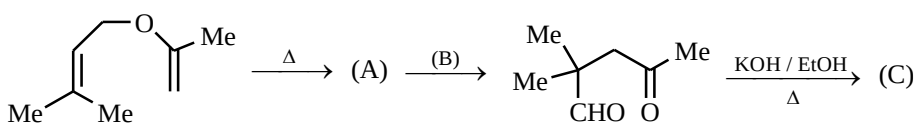
PART – A: (Multi Correct Answer Type)

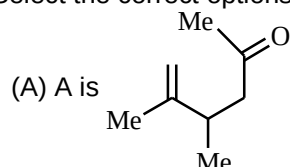
This section contains **06 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. The alcohol  on dehydration gave three isomeric alkenes A, B and C. One mole of each of these are subjected to ozonolysis. Then A gave one mole of a dialdehyde B gave one mole of an aldehyde and one mole of ketone where as C gave one mole of keto-aldehyde. The correct statements from the following are:
- (A) in the formation of A from the reactant ring expansion takes place
 (B) in the formation of B from the reactant 1, 2 hydride shift takes place
 (C) B is Hoffmann's product
 (D) C is Zaitsev's product

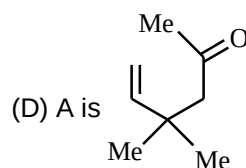
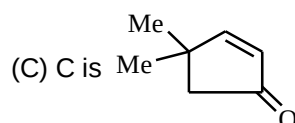
2. Which of the following shall be quantitatively reduced when HCHO is mixed and solution made strongly alkaline:



3. 
 Select the correct options:

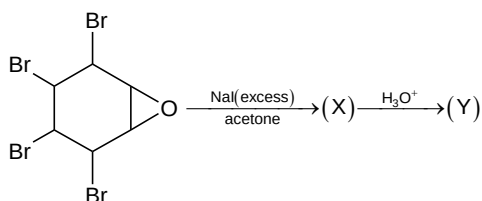


(B) B is O₃ followed by (CH₃)₂S



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4.

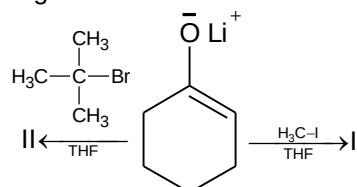


Select the correct statements regarding the final product (Y):

- (A) It can be identified with neutral FeCl_3 solution
- (B) On treatment with Zn-dust, it yields a product having degree of unsaturation = 4
- (C) It undergoes coupling reaction with benzene diazonium chloride at low temperature ($0 - 5^\circ\text{C}$) in slightly basic medium
- (D) It liberates H_2 gas with Na

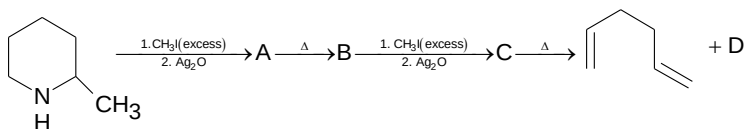
5.

The lithium enolate base from cyclohexanone reacts with alkyl halides, often in different ways. As shown here, methyl iodide and *tert*-butyl bromide react to give different organic products, I and II, together with lithium halides. What are the products from these reactions?



- (A) I is 1-methoxycyclohexene
- (B) II is 1-*t*-butoxycyclohexene
- (C) I is 2-methylcyclohexanone
- (D) II is a mixture of cyclohexanone and 2-methylpropene

6.



Compound 'B' can not be in the above sequence is:

- (A)
- (B)
- (C)
- (D) $(\text{CH}_3)_3\text{N}$

Space for rough work

PART – A: (Matrix-Match Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO (02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Match the reactant and product in Column – I with their corresponding reagents in Column – II:

List – I (Reactant and product)		List – II (Reagents)	
(I)		(P)	$\text{H}_2\text{O}_2 + \text{OH}^- / \text{H}_3\text{O}^+$
(II)		(Q)	$\text{Zn} - \text{Hg} + \text{conc. HCl}$
(III)		(R)	$\text{K}_2\text{S}_2\text{O}_8 + \text{OH}^- / \text{H}_3\text{O}^+$
(IV)		(S)	$\text{NH}_2 - \text{NH}_2 / \text{OH}^-$
		(T)	Ag_2O (moist)
		(U)	Na / Ether

7. Which of the following options has the correct combination considering List-I and List-II?

Options

(A) (I) $\rightarrow$ (P), (II) $\rightarrow$ (R)

(B) (II) $\rightarrow$ (R), (III) $\rightarrow$ (Q)

(C) (III) $\rightarrow$ (S), (IV) $\rightarrow$ (R)

(D) (IV) $\rightarrow$ (U), (I) $\rightarrow$ (R)

8. Which of the following options has the correct combination considering List-I and List-II?

Options

(A) (I) $\rightarrow$ (R), (II) $\rightarrow$ (P)

(B) (III) $\rightarrow$ (S), (IV) $\rightarrow$ (Q)

(C) (III) $\rightarrow$ (R), (IV) $\rightarrow$ (U)

(D) (IV) $\rightarrow$ (R), (I) $\rightarrow$ (Q)

Space for rough work

Match List I with List II and select the correct options given below:

List – I		List – II	
(I)	$(\text{COOH})_2 \cdot 2\text{H}_2\text{O} \xrightarrow[150^\circ\text{C}]{\Delta} (\text{I})$	(P)	$2\text{CO}_2 + \text{H}_2\text{O}$
(II)	$(\text{I}) \xrightarrow[\text{H}_2\text{SO}_4, \text{ at } 90^\circ\text{C}]{200^\circ\text{C}}$	(Q)	$(\text{COOH})_2$
(III)	$\begin{array}{c} \text{COOH} \\ \diagdown \\ \text{C} \\ \diagup \\ \text{COOH} \end{array} \xrightarrow[\Delta]{\text{P}_2\text{O}_5}$	(R)	$\text{CO} + \text{CO}_2 + \text{H}_2\text{O}$
(IV)	$(\text{I}) \xrightarrow[\text{KMnO}_4]{\Delta}$	(S)	Carbon sub oxide $\text{O} = \text{C} = \text{C} = \text{C} = \text{O}$
		(T)	$\text{CO} + \text{H}_2\text{O}$
		(U)	$\text{O} = \text{C} = \text{C} = \text{O}$

Codes:

	1	2	3	4
(A)	3	1	2	4
(B)	2	1	4	3
(C)	Q	R	S	P
(D)	4	1	2	3

9. Which of the following options has the correct combination considering List-I and List-II?

Options

- (A) (I) $\rightarrow$ (P), (II) $\rightarrow$ (R) (B) (II) $\rightarrow$ (R), (III) $\rightarrow$ (T)
 (C) (I) $\rightarrow$ (Q), (II) $\rightarrow$ (R) (D) (IV) $\rightarrow$ (S), (I) $\rightarrow$ (U)

10. Which of the following options has the correct combination considering List-I and List-II?

Options

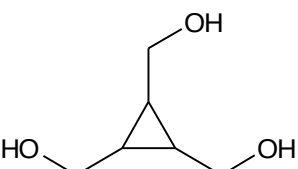
- (A) (I) $\rightarrow$ (U), (II) $\rightarrow$ (S) (B) (III) $\rightarrow$ (P), (IV) $\rightarrow$ (Q)
 (C) (I) $\rightarrow$ (R), (IV) $\rightarrow$ (T) (D) (III) $\rightarrow$ (S), (IV) $\rightarrow$ (P)

Space for rough work

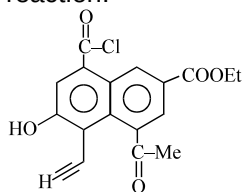
PART – C: (Numerical Answer Type)

This section contains **08 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer and **there will be no negative marking**.

1. When 3-methyl pentane is subjected for monohalogenation using $\text{Cl}_2/h\nu$, a total number chiral monchloro product is/are
2. 4.6 g of a polyhydric alcohol was treated with an excess of methyl magnesium bromide to produce 3.36 litre of CH_4 at STP. Calculate number of $-\text{OH}$ group present in the alcohol.
(molecular weight of alcohol = 92)

3.  $\xrightarrow[\Delta]{\text{conc. H}_2\text{SO}_4}$ (X) (Major product)
Degree of unsaturation a molecule of 'X' is

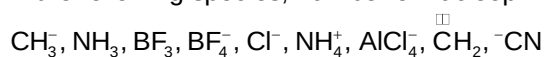
4. How many moles of Grignard reagent can react with one mole of following compound for complete reaction.



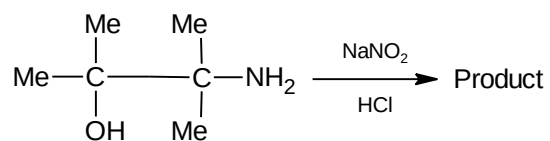
5. The number of chiral compounds produced upon monochlorination of 2-methylbutane is ____

6. How many compounds from the following list are isomers of D – glucose?
D-Mannose, D-Fructose, D-Gulose, D-Idose, D-Galactose, D-Arabinose, D-Ribose.

7. In the following species, number of nucleophiles is ____



8.



The number of methyl groups attached to SP^2 hybridized carbon in product is ____.